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a-Nventory

an Automated Inventory Management System
for IT assets managed by the College of Computing* Technical Team

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A. HIGHLIGHTS

Question: What problems do equipment borrowers and managers face with the manual logbook, and how strong is the support for an automated system?

Role & Contributions: I designed the survey questionnaires to gather requirements, collected 27 responses (20 faculty, 3 technical team members, 4 student assistants), coded the checkbox and open-ended answers, and built the summary tables and charts in Google Sheets. Based on the results, I designed and proposed an automated inventory management system and wrote the analysis, key findings, and limitations.

Main Findings:

1. Lost items and lack of accountability for borrowed items were the shared problems (14 and 13 of 27 respondents, 52% and 48%).
2. Faculty's top issue was availability (12 of 20, 60%). Staff's top issues were lost items and accountability for borrowed items (6 of 7 each).
3. All 27 respondents agreed the current process is inefficient.
4. Need and willingness were rated very high: a mean necessity of 8.93 out of 10 (no rating below 6), and all 27 rated adoption likelihood 5 out of 5.

Overall, the findings support further development and evaluation of an automated inventory system as a proposed solution to the identified inventory management challenges. However, these results reflect respondents' perceptions and stated intentions; the system's actual effectiveness and adoption would need to be evaluated after implementation.

B. PROJECT DESCRIPTION

1. Description

a-Nventory is an automated inventory management system to streamline the borrowing, returning, and record-keeping of electrical items managed by the technical team of the College of Computing. By supporting both mobile and PC-based QR code scanning, the system will track users, transaction dates, and item identification. Additionally, it will allow reservations, send automatic notification alerts, and provide an intuitive user interface for monitoring transactions, available, and reserved equipment.

2. Key Features

a. Dual QR Code Scanning Support – Works on both mobile devices and PC-connected scanners for scanning user IDs and item QR codes
b. Reservation System – Users can reserve items through the system
c. Transaction and Inventory UI – Users can view transaction history and check available or reserved items
d. Custom and smart alert notifications:
    - For borrowers: Return reminders and overdue alerts
    - For admins: Due returns, overdue items, and repair needs
e. Admins can send messages to borrowers for reminders or follow-up.

*The college name has been generalized to protect institutional privacy.

C. METHOD

3. Respondents

The respondents are the faculty of the said college, the technical team, and the student assistants who facilitate the borrowing and returning processes, and the logbook maintenance. Respondents (n = 27):

Table 3.1 Respondents

Group	Respondents	Share
Faculty (borrowers)	20	74%
Technical team (equipment managers)	3	11%
Student assistants (logbook keepers)	4	15%
Total	27	100%

4. Questionnaires

Faculty answered 4 questions (one open-ended question about challenges, plus an inefficiency question, a 1 to 10 necessity rating, and a 1 to 5 adoption-likelihood rating). The technical team and student assistants answered 8 questions, including a problem checklist, problem frequency, who may borrow, and restocking frequency.

Important difference: Faculty answered an open-ended question that I coded into categories afterward. The technical team and student assistants chose from a fixed checklist that did not include "no easy way to check availability" or, for faculty, "incomplete records." Counts for those categories are therefore not comparable across groups, and I compare groups only on categories all three could report.

NOTE: Open-ended responses were paraphrased and grouped by theme after collection.

4.1. Questionnaire for Faculty Members

Proposed Project: ***a-Nventory*** - an automated inventory management system for IT assets managed by the College of Computing technical team

Instructions: Please read the project overview and answer the questions below.
Project Overview:
The College of Computing technical team manages IT assets manually, leading them to potential issues in tracking items. a-Nventory aims to solve these problems through QR scanning and real-time tracking.

Questions:

1. Do you agree that the current inventory management process is inefficient? (Disagree, Neutral, Agree) _____
2. If you have borrowed items such as projectors or cables in the past from the team, what are the main challenges you've observed in the current system?
   _____
   _____
3. Please rate the necessity of an automated system from 1 to 10 (1 = Not Needed, 10 = Highly Needed) _____
4. From a scale of 1 - 5, how likely are you to adopt an automated inventory management system if it were implemented in your organization? (1: Not likely at all, 3: Neutral, 5: Very likely) _____

Department: _____

4.2. Questionnaire for the Technical Team and Student Assistants

a-Nventory: Streamlining the borrowing, returning, and inventory of IT assets managed by the College of Computing technical team

Good day! Thank you for participating in the survey. Your responses will remain confidential and will not be used outside this project.

1. How do you currently track IT assets inventory?
   _____
2. What problems do you face with the current inventory process? (Check all that apply)
   - ☐ Difficulty tracking borrowed equipment
   - ☐ Misplaced or lost items
   - ☐ Lack of accountability for borrowed items
   - ☐ Others (please specify): _____
3. How often do you experience these problems?
   - ☐ Daily
   - ☐ Weekly
   - ☐ Monthly
   - ☐ Other (please specify): _____
4. Who is authorized to borrow these assets? (Check all that apply)
   - ☐ College of Computing Students
   - ☐ College of Computing Faculty Members
   - ☐ College of Computing Staffs
   - ☐ Students from other colleges
   - ☐ Faculty Members of other Colleges
   - ☐ Staffs of other Colleges
   - ☐ Other (please specify): _____
5. How often do you restock IT supplies/equipment?
   - ☐ Weekly
   - ☐ Monthly
   - ☐ Annually
   - ☐ Rarely/Only if needed
   - ☐ Other (please specify): _____
6. Do you agree that the current inventory management process is inefficient? (Disagree, Neutral, Agree) _____
7. Rate the necessity of an automated inventory system from 1 - 10 scale (1: Not needed, 10: Highly Needed) _____
8. From a scale of 1 - 5, how likely are you to adopt an automated inventory management system if it were implemented in your organization? (1: Not likely at all, 3: Neutral, 5: Very likely) _____

Name (Optional): _____
Designation (Required): _____

D. DATA: SURVEY RESULTS & VISUALIZATION

5. Survey Results

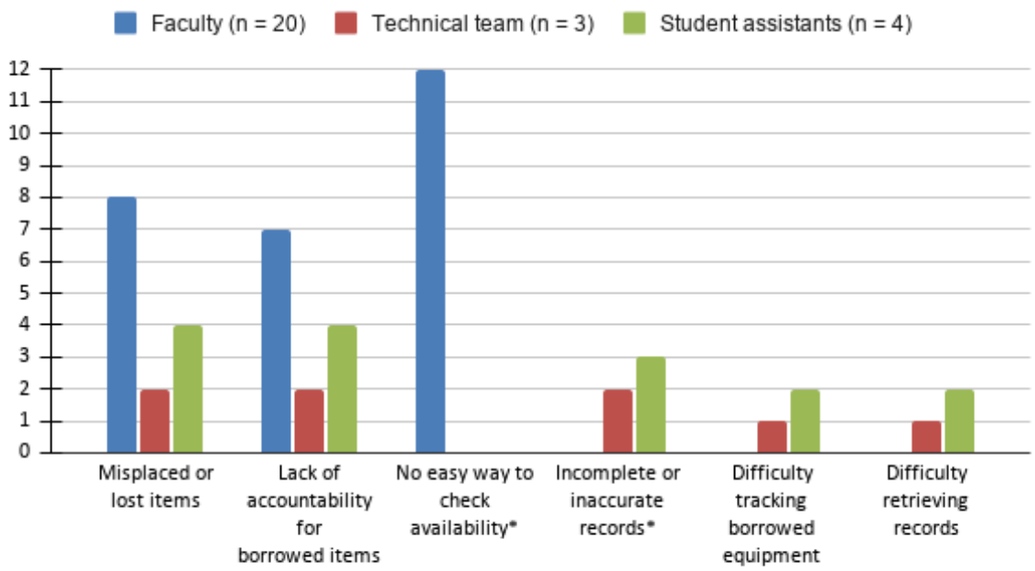
View the complete survey results: **a-Nventory Survey Results**

Table 5.1 Problems with the current logbook

Problem	Faculty (n = 20)	Technical team (n = 3)	Student assistants (n = 4)	All (n = 27)	% of all respondents
Misplaced or lost items	8	2	4	**14**	**52%**
Lack of accountability for borrowed items	7	2	4	**13**	**48%**
No easy way to check availability*	12	0	0	**12**	**44%**
Incomplete or inaccurate records*	0	2	3	**5**	**19%**
Difficulty tracking borrowed equipment	0	1	2	**3**	**11%**
Difficulty retrieving records	0	1	2	**3**	**11%**

**Not offered to every group (see Questionnaires), so zeros do not mean the problem is absent.*

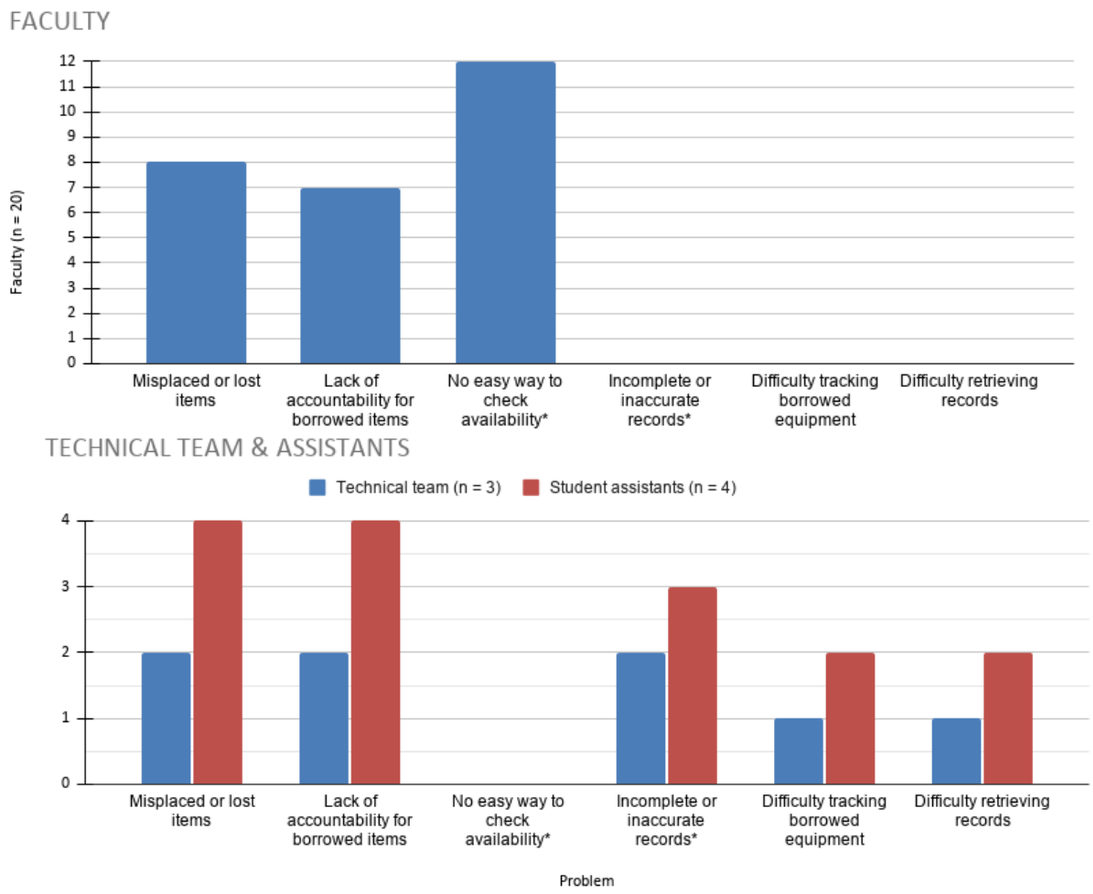
Figure 5.1 Encountered Problems of All Respondent Groups



Misplaced or lost items and accountability are the shared problems. They were the top problems for the technical team and assistants (6 of 7 staff reported each, 86%) and among the top three for faculty. The faculty's biggest issue is that there is no easy way to check availability of items. Staff (technical team and assistants) are the ones feeling the record-keeping problems. 5 of 7 staff (71%) reported incomplete or inaccurate records. Because the assistants maintain the logbook, this is a firsthand view.

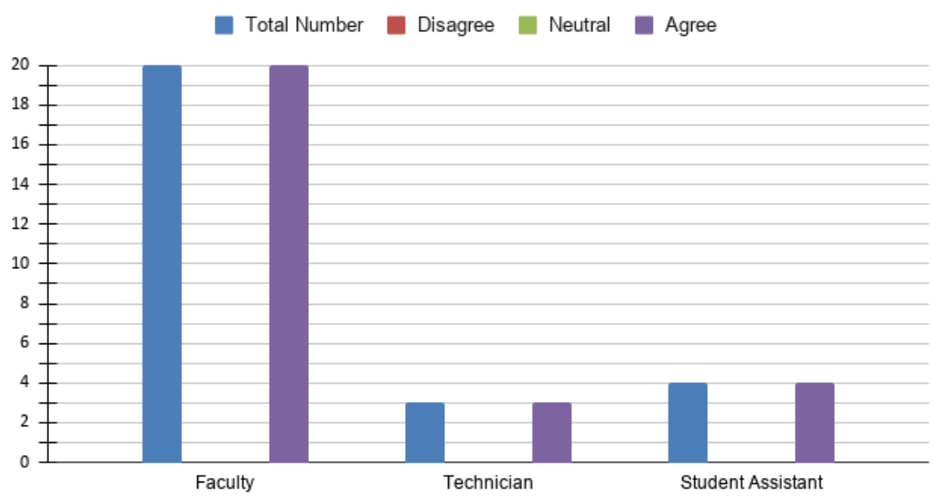
These issues highlight the need for more reliable transaction records and borrower tracking. The proposed system could address these concerns through QR code scanning and transaction history monitoring.

Figure 5.2 Encountered Problems of the Faculty Members (Borrowers) VS Technical Team and Assistants



Faculty members most frequently reported difficulty checking equipment availability, followed by misplaced or lost items and lack of accountability for borrowed items. The technical team reported misplaced or lost items, lack of accountability, and incomplete or inaccurate records. Student assistants reported misplaced or lost items and lack of accountability most frequently. Because student assistants maintain the manual logbook, these findings suggest that a more structured recording process could help them track borrowing and returning transactions more consistently. These findings suggest that improved visibility into item availability and more consistent borrowing records could help address faculty members' concerns and a centralized system with transaction tracking could help improve record consistency.

Figure 5.3 Response to “Do you agree that the current system is inefficient?”



All 27 respondents (100%) agreed that the current inventory management process is inefficient. This consistent response across the respondent groups indicates a clear perceived need to improve the existing process.

Table 5.2 Necessity of an Automated Inventory System (AIMS) Rating

Rating	**6**	**7**	**8**	**9**	**10**
Faculty	2	3	3	5	7
Technical Team	0	0	0	1	2
Student Assistants	0	0	0	0	4
All	**2**	**3**	**3**	**6**	**13**

Respondents generally rated the proposed automated inventory system as highly necessary. The mean necessity rating was 8.93 out of 10. Out of 27 respondents, 13 (48%) gave it the maximum rating of 10, 22% gave it 9, and no one rated it below 6. These results indicate strong perceived support for the proposed system.

Table 5.3 Willingness to Adopt to the AIMS if implemented

Rating	**1**	**2**	**3**	**4**	**5**
Faculty	0	0	0	0	20
Technical Team	0	0	0	0	3
Student Assistants	0	0	0	0	4
All	**0**	**0**	**0**	**0**	**27**

All 27 respondents (100%) indicated that they would be willing to adopt the proposed system. This suggests a high level of stated willingness to adopt the system if implemented.

5.1 Open-ended Responses: Themes in Reported Challenges

Faculty were asked what challenges they had observed when borrowing equipment, and the technical team and student assistants described challenges in their own words. Each response was paraphrased and grouped by theme. A respondent could mention more than one theme and is counted once under each, so counts add up to more than the number of respondents.

Table 5.4 Themes in Faculty Responses

Theme	**Faculty (n = 20)**	**% of Faculty**	**Example**
Availability or item status unclear	12	60%	Can’t tell whether an item is free or current without visiting or contacting someone
Can’t check reservations in advance	4	20%	No way to see reservations beforehand
Recording depends on someone being present	3	15%	A transaction is logged only if staff are there
Missing accessories on return	2	10%	A projector comes back without its cable
Hard to trace a borrowed item	2	15%	The borrower is unknown when an entry is missing
Incomplete records when busy	2	10%	Entries are missed when the station is crowded

Table 5.5 Themes in Technical Team and Assistants Responses

Theme	Technical Team (n = 3)	Student Assistants (n = 4)	All staff (n = 7)	% of Staff
Accessories missing on return	0	3	3	43%
Last borrower hard to identify when the entry is incomplete	1	2	3	43%
Transactions not recorded when many people need equipment	2	0	2	29%
Overdue follow-up slowed by incomplete borrower details	0	2	2	29%
Equipment collected when no assistant is at the station	0	1	1	14%

Availability was the dominant faculty concern: 12 of 20 faculty (60%) described not knowing whether an item was free, and 4 more (20%) mentioned being unable to check reservations in advance. Gaps in recording appeared in both groups. Faculty described records depending on staff being present (3 of 20) or being incomplete when the station was busy (2 of 20), and staff described difficulty identifying the last borrower (3 of 7) and transactions going unrecorded when many people needed equipment (2 of 7). Missing accessories, such as a cable returned without its projector, came up in 5 of 27 responses (2 faculty, 3 student assistants), which suggests tracking accessories together with each item.

E. LIMITATIONS

1. Near-unanimous answers. Everyone agreeing and rating the system highly may reflect that the survey introduction described the current process negatively, that respondents knew me, or that people tend to agree with proposals. It does not tell us how many people would actually use the system.
2. Open-ended responses were paraphrased and grouped by theme, with a respondent counted once under each theme mentioned. These theme counts are a separate grouping from Table 5.1 and are not expected to match it.
3. Stated intentions, not behavior. Adoption and effectiveness can only be tested after implementation.

F. KEY FINDINGS

1. Lost items and lack of accountability are the shared problems. They were reported by 14 and 13 of 27 respondents (52% and 48%), including 6 of 7 staff each.
2. Faculty’s top issue is availability. 12 of 20 faculty (60%) said there is no easy way to check what is available. Staff’s top issues were lost items and accountability (6 of 7 each), and 5 of 7 reported incomplete or inaccurate records.
3. All 27 respondents agreed the current process is inefficient.
4. Need and willingness were rated very high: a mean necessity of 8.93 out of 10 (no rating below 6), and all 27 rated adoption likelihood 5 out of 5.

Overall, the findings support further development and evaluation of a-Nventory as a proposed solution to the identified inventory management challenges. However, these results reflect respondents' perceptions and stated intentions; the system's actual effectiveness and adoption would need to be evaluated after implementation.